

```
In [ ]: # Exp_7
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        # Roll no:16
        # Sec:A
        # SubjeCT:ET1
        # Date:22/09/2025
```

```
In [1]: #Aim: Linear regression
```

```
In [2]: import pandas as pd
```

```
In [4]: import os
```

```
In [5]: os.getcwd()
```

```
Out[5]: 'C:\\Users\\user'
```

```
In [6]: os.chdir("C:\\Users\\USER\\Desktop")
```

```
In [13]: data=pd.read_csv("C:\\Users\\user\\Desktop\\data.salary.csv")
```

```
In [14]: data.head(20)
```

```
Out[14]:
```

	YearsExperience	Salary
0	1.1	39343
1	1.3	46205
2	1.5	37731
3	2.0	43525
4	2.2	39891
5	2.9	56642
6	3.0	60150
7	3.2	54445
8	3.2	64445
9	3.7	57189
10	3.9	63218
11	4.0	55794
12	4.0	56957
13	4.1	57081
14	4.5	61111
15	4.9	67938

	YearsExperience	Salary
16	5.1	66029
17	5.3	83088
18	5.9	81363
19	6.0	93940

In [15]: `data.tail()`

Out[15]:

	YearsExperience	Salary
25	9.0	105582
26	9.5	116969
27	9.6	112635
28	10.3	122391
29	10.5	121872

In [16]: `data.describe()`

Out[16]:

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

In [17]: `data.shape`

Out[17]: (30, 2)

In [18]: `data.size`

Out[18]: 60

In [19]: `data.ndim`

Out[19]: 2

In [20]: `data.columns`

```
Out[20]: Index(['YearsExperience', 'Salary'], dtype='object')
```

```
In [21]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column          Non-Null Count  Dtype  
---  -
 0   YearsExperience  30 non-null    float64
 1   Salary          30 non-null    int64  
dtypes: float64(1), int64(1)
memory usage: 608.0 bytes
```

```
In [22]: data.isnull().sum()
```

```
Out[22]: YearsExperience    0
Salary                    0
dtype: int64
```

Independent and dependent variables

```
In [23]: X=data.drop('Salary',axis=1)
```

```
In [24]: X.head()
```

```
Out[24]:   YearsExperience
0          1.1
1          1.3
2          1.5
3          2.0
4          2.2
```

```
In [26]: y=data.Salary
```

```
In [27]: y.head()
```

```
Out[27]: 0    39343
1    46205
2    37731
3    43525
4    39891
Name: Salary, dtype: int64
```

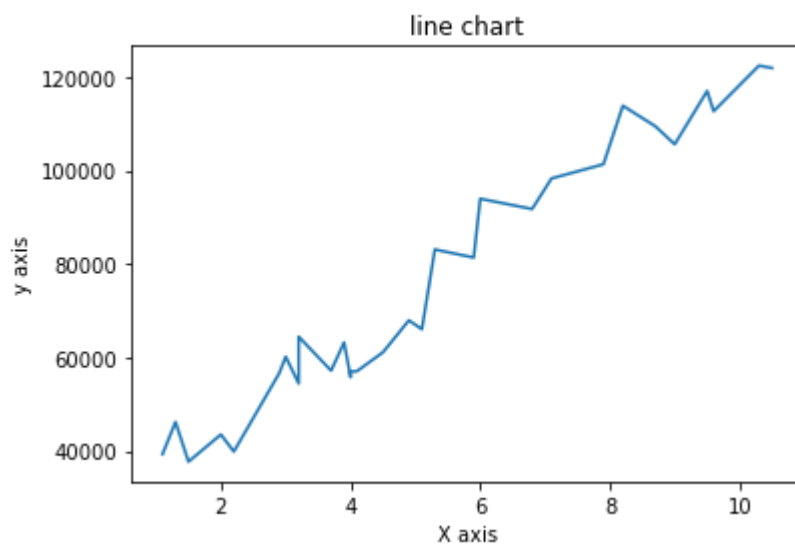
line chart

```
In [28]: from matplotlib import pyplot as plt
```

In [29]:

```
plt.plot(X,y)
plt.title("line chart")
plt.xlabel("X axis")

plt.ylabel("y axis")
plt.show()
```



In []: